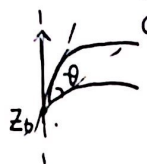


解析函数导数的几何意义:



$$C_1: z = z_1(t), \quad a \leq t \leq b, \quad z_1'(t) \neq 0, \quad \forall t \in (a, b).$$

$$C_2: z = z_2(t), \quad z_2'(t) \neq 0.$$

$$\text{在 } z_0 \text{ 处: } z_1(t_0) = z_2(t_0) = z_0.$$

$$\arg \frac{z_2'(t_0)}{z_1'(t_0)} = \theta, \text{ 即曲线 } C_1 \text{ 与 } C_2 \text{ 之间的夹角.}$$

$$\text{这意味着: } \arg z_2'(t_0) - \arg z_1'(t_0) = \theta \in [0, \pi).$$

在此基础上, 考虑解析映射:

$$w(t) = f(z(t)), \quad w'(t) = f'(z(t))z'(t), \quad \arg w'(t) = \arg f'(z(t)) + \arg z'(t).$$

保角性: 映射后 C_1, C_2 的夹角大小与方向不变.

$$\varphi = \arg \frac{w_2'(t_0)}{w_1'(t_0)} = \arg \frac{f'(z_2(t_0))z_2'(t_0)}{f'(z_1(t_0))z_1'(t_0)} = \arg \frac{z_2'(t_0)}{z_1'(t_0)} = \theta.$$

* 任何一个不为常数的可导函数几乎是处处保角的.

↳ 例如 $\sin x, e^x, \ln x$ 等.

$$\text{分式线性函数: } w = f(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}.$$

$$f'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0, \quad (ad - bc \neq 0).$$

$$\hookrightarrow \det(A) = ad - bc \neq 0. \quad (A \text{ 是可逆的 } 2 \times 2 \text{ 复常数矩阵})$$

$$\text{若 } ad - bc = 0, \quad f'(z) \equiv 0. \text{ 那么 } f(z) \text{ 是恒等于 } f(0) = \frac{b}{d}. \text{ (不考).}$$

因此, 分式线性映射是处处保角的 (非常数时).

对于矩阵 A , A 与 λA 对应的分形映射是一样的.

$$\text{总可以取 } \lambda = \frac{\pm 1}{\sqrt{ad - bc}}, \text{ 使得 } \det(\lambda A) = 1.$$

$$\text{若 } z' = \frac{a_1 z + b_1}{c_1 z + d_1}, \quad w = \frac{a_2 z' + b_2}{c_2 z' + d_2}, \text{ 总会有 } w = \frac{Az + B}{Cz + D}.$$

$$\text{其中 } AD - BC = (ad_1 - b_1c_1) \times (a_2d_2 - b_2c_2) \neq 0.$$

$$\text{若 } \det(\vec{A}) = 1, \quad \det(\vec{B}) = 1.$$

分形映射的叠加仍是分形映射.

$$\text{若 } w = \frac{az + b}{cz + d}, \text{ 反解出 } z = \frac{-dw + b}{cw - a} \quad (\text{假设 } \det \neq 0).$$

分式线性映射可逆

$$\text{(单位恒等映射)} \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{不动点} = f(z) = z$$

1. 没有不动点.

$$\text{平移: } w = z + \alpha, \quad \alpha = a + bi \neq 0, \quad a, b \in \mathbb{R}, \quad A = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$$

2.

$$\text{伸缩: } w = az, \quad a \in \mathbb{C}, \quad a \neq 0.$$

$$\text{不妨写 } a = re^{i\theta} = |a|e^{i\theta}.$$

$$\text{那么: } w = |a|e^{i\theta}z$$

伸缩 $\uparrow$ $\uparrow$ 坐标旋转.

$$A = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$$

3. 倒数(映射). $w = \frac{1}{z}$

对应 $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ 有两个不动点 ± 1 .

$c \neq 0$ 时, 把有限映为无穷远

$\Delta \Gamma \times$ 圆 = 直线 $\cup$ 圆.

定理 = 分式^{线性}映射保证广义圆及其对称点.

定义 = 对称点 (关于直线的)

圆周的对称点 (方向不变, 距离取倒数). 即 $z' - z_0 = \lambda(z - z_0)$, 且 $|z' - z_0| |z - z_0| = r^2$. ($\lambda > 0$).

故代入得到对称点 $z' = z_0 + \frac{r^2}{|z - z_0|^2} (z - z_0)$

$= z_0 + \frac{r^2}{(z - z_0)(\bar{z} - \bar{z}_0)}$ (拆成共轭后约去)

统一的定义: (对 $\Gamma \times$ 圆).

$$z' = \frac{\bar{z}z_0 - \bar{z}_0z + r^2}{\bar{z} - \bar{z}_0}$$

特别地, 对单位圆的对称点: $z' = \frac{1}{\bar{z}}$

→ 分式线性映射可以拆成平移、旋转、倒数.

例 1. 对于 $w = \frac{az+b}{cz+d}$, a, b, c, d 满足什么条件时, w 将单位圆周映成直线

答 条件为: $|c| = |d|$.

必要性:

由于分式线性映射保证广义圆.

存在 $|z_0| = 1$ 使 $cz_0 + d = 0 \Leftrightarrow |cz_0| = |d| \Leftrightarrow |c| = |d|$.

充分性:

$$|c| = |d| \Rightarrow c = de^{i\theta_0} \Rightarrow cz + d = de^{i\theta_0}z + d = d(e^{i\theta_0}z + 1).$$

当 $z = e^{-i\theta_0}$ 在圆周上时, $cz + d = 0$. (这时 $|z| = 1$).

故圆周上一点映为无穷远. 由于分式线性映射保广义圆, 故得证.

Δ 分式线性映射:

$$w = \frac{az+b}{cz+d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc \neq 0.$$

1x 不动点 1) 平移 = $w = z - z_1 + w_1 = z + \alpha$. (其中 $\alpha = w_1 - z_1$) 此时 $z_1 \rightarrow w_1$.

2x 不动点 2) 线性 = (z_1, z_2 互异) = $w = \frac{z - z_1}{z_2 - z_1} w_1$ (但 w_1, w_2 可以相同)

$$w = \frac{w_2 - w_1}{z_2 - z_1} (z - z_1) + w_1 = \alpha z + \beta. \quad (\text{此时 } z_1 \rightarrow w_1, z_2 \rightarrow w_2)$$

3x 不动点 3) $z_k \rightarrow w_k = (k=1, 2, 3)$. (z_k 互异) (但 w_k 可以相同)

$$\text{满足: } \frac{w - w_1}{w - w_2} / \frac{w_3 - w_1}{w_3 - w_2} = \frac{z - z_1}{z - z_2} / \frac{z_3 - z_1}{z_3 - z_2} \quad \text{不变式 I.}$$

$n \times$ 不动点 4) $z_k \rightarrow w_k = (k=1, 2, 3, 4, \dots)$. (z_k 互异)

可能会不满足.

因此, 利用拉格朗日插值公式 (LIP)

$$l_k(x) = \frac{(x-x_1)(x-x_2)\cdots(x-x_{k-1})(x-x_{k+1})\cdots(x-x_n)}{(x_k-x_1)\cdots(x_k-x_{k-1})(x_k-x_{k+1})\cdots(x_k-x_n)}$$

实数下:

次数为 $n-1$. 当且仅当 $k=j$ 时, $l_k(x_j) = 1$ 时, 否则为 0.

$$\text{再令 } L_n(x) = \sum_{k=1}^n y_k l_k(x), \quad k=1, 2, \dots, n.$$

可证 $L_n(x)$ 的次数为 $n-1$.

复数下:

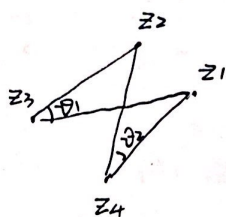
令 $z_k = x_k, w_k = y_k, z = x, w = y$.

$n-1$ 次多项式的映射

例 2. 若 z_1, z_2, z_3, z_4 任三点不在一直线上, 试求它们是否在同一圆上.

(法一 = 初等方法、暴力代入解方程)(可忽略).

法二:



只需证 $\angle \theta_1 = \theta_2$,

$$\arg \frac{z_3 - z_1}{z_3 - z_2} = \arg \frac{z_4 - z_1}{z_4 - z_2}.$$

可以写成:

$$\arg \left(\frac{z_3 - z_1}{z_3 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_1} \right) = 0 \text{ (或为 } \pi \text{)}$$

$$\text{故等价于 } \frac{z_3 - z_1}{z_3 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_1} \in \mathbb{R}.$$

(> 0)

此时, z_1, z_2, z_3, z_4 共圆 (充要条件)

~~法三:~~

例 3. 把单位圆映射为另一个圆盘. 把圆盘中的 z_1 映为 0.

$$w = \alpha \frac{z - z_1}{z - \bar{z}_1} \quad \text{其中 } z_1 \text{ 与 } \bar{z}_1 \text{ 为对称点, } z_1' = \frac{1}{\bar{z}_1}$$

$$\text{故 } w = \alpha \frac{z - z_1}{z - \frac{1}{\bar{z}_1}} = \alpha (z - z_1) \frac{\bar{z}_1}{1 - \bar{z}_1 z} = \beta \frac{z - z_1}{1 - \bar{z}_1 z}.$$

$$\text{而又因为 } w(z) = \beta \frac{ad - bc}{(1 - \bar{z}_1 z)^2}, \quad (a, b, c, d \in \mathbb{C}).$$

$$\text{故 } w'(z) = \beta \frac{1 - |z_1|^2}{(1 - \bar{z}_1 z)^2}$$

由于 w 是解析函数, 且满足最大模原理:

$$|z| = 1 \Leftrightarrow |w| = 1$$

当 $|w| = 1$ 时, $|z| = 1$.

$$\text{即 } 1 = |\beta| \left| \frac{z - z_1}{1 - \bar{z}_1 z} \right|_{|z|=1}$$

由于 $1 = z\bar{z}$.

$$\text{故 } 1 = |\beta| \left| \frac{z - z_1}{z(\bar{z} - \bar{z}_1)} \right| = \frac{|\beta|}{|z|} \left| \frac{z - z_1}{\bar{z} - \bar{z}_1} \right| \quad |z| < 1, |z| = 1 \Rightarrow z = \bar{z}_1$$

$$\text{故 } 1 = |\beta| \cdot 1 \quad \text{即 } |\beta| = 1 \Rightarrow \beta = e^{i\theta}.$$

$$\text{那么, } w = e^{i\theta} \frac{z - z_1}{1 - \bar{z}_1 z}, \quad \text{其中 } |z_1| < 1, \theta \in [0, 2\pi).$$

(考点 = 单位映射(圆)到另一圆的一角形式)

例 4. (例 3 的变式) 由 Γ 圆心为 z_0 映到 Γ 圆心为 w_0 的圆盘, 其中把 $z_1 \rightarrow w_0$.

$$\text{作 } z' = \frac{z-z_0}{r}, \quad w' = \frac{w-w_0}{R}.$$

这时把 $z' \rightarrow w'$, 把 z' 映为 w' .

$$w' = e^{i\theta} \frac{z' - z'_1}{1 - \bar{z}'_1 z'}$$

$$\text{再代入, } \frac{w-w_0}{R} = e^{i\theta} \frac{\frac{z-z_0}{r} - \frac{z-z_1}{r}}{1 - \frac{\bar{z}-\bar{z}_0}{r} \cdot \frac{z-z_1}{r}}$$

$$\text{那么 } w = w_0 + R e^{i\theta} \frac{z-z_1}{r^2 - (\bar{z}_1 - \bar{z}_0)(z-z_0)}, \text{ 其中 } |z_1 - z_0| < r, \theta \in [0, 2\pi)$$

再化简, 令 $w_0 = 0, z_0 = 0, R = r = 1$.

$$\text{又可以得到 } w = e^{i\theta} \frac{z-z_1}{1 - \bar{z}_1 z}, \text{ 其中 } |z_1| < 1, \theta \in [0, 2\pi)$$

例 1: $\Downarrow$

$$\text{不变式 II: } \frac{|dw|}{1-|w|^2} = \frac{|dz|}{1-|z|^2}$$

$$\Leftrightarrow \left| \frac{dw}{dz} \right| (1-|z|^2) = 1-|w|^2.$$

$$\text{而 } \frac{dw}{dz} = \frac{ad-bc}{ccz+d}, \text{ 此处 } a=e^{i\theta}, b=-z_1 e^{i\theta}, c=-\bar{z}_1, d=1.$$

$$\text{代入, } \frac{dw}{dz} = \frac{e^{i\theta}(1-|z_1|^2)}{(1-\bar{z}_1 z)^2},$$

$$\text{而 } \left| \frac{dw}{dz} \right| = \frac{1-|z_1|^2}{|1-\bar{z}_1 z|^2} > 0.$$

得到 $1-|z|^2$ 与 $1-|w|^2$ 同号.

圆内到圆内、圆上到圆上、圆外到圆外.

不变式 II 的证明:

$$\text{由于 } \dots \Leftrightarrow \left| \frac{dw}{dz} \right| (1-|z|^2) = 1-|w|^2 \Leftrightarrow \frac{1-|z_1|^2}{|1-\bar{z}_1 z|^2} (1-|z|^2) = 1-|w|^2.$$

只需证右式即可.

$$\text{而 } 1-|w|^2 = 1-w\bar{w} = 1 - e^{i\theta} \frac{z-z_1}{1-\bar{z}_1 z} \cdot \overline{e^{i\theta} \frac{z-z_1}{1-\bar{z}_1 z}}$$

$$= 1 - e^{i\theta} \frac{z-z_1}{1-\bar{z}_1 z} e^{-i\theta} \frac{\bar{z}-\bar{z}_1}{1-\bar{z}_1 \bar{z}}$$

$$= 1 - \frac{(z-z_1)(\bar{z}-\bar{z}_1)}{|1-\bar{z}_1 z|^2} = \frac{(1-\bar{z}_1 z_0)(1-\bar{z}_1 \bar{z}) - (z-z_1)(\bar{z}-\bar{z}_1)}{|1-\bar{z}_1 z|^2}$$

$$= \frac{1 - \bar{z}_1 z - \bar{z}_1 \bar{z} + \bar{z}_1 z_1 \bar{z} - z \bar{z} + \bar{z}_1 z + \bar{z}_1 \bar{z} - z_1 \bar{z}_1}{|1-\bar{z}_1 z|^2}$$

$$= \frac{1 - z_1 \bar{z} - z \bar{z} + |z_1|^2 |z|^2}{|1-\bar{z}_1 z|^2} = \frac{(1-|z_1|^2)(1-|z|^2)}{|1-\bar{z}_1 z|^2}.$$

右式即证.

对于更一般的:

$$w = w_0 + e^{i\theta} R \frac{z - z_1}{r^2 - (\bar{z}_1 - \bar{z}_0)(z - z_0)} \quad \text{其中 } |z_1 - z_0| < r, \theta \in [0, 2\pi)$$

$$\text{令 } w' = \frac{w - w_0}{R}, \quad z' = \frac{z - z_0}{r}$$

$$\text{由 } \frac{|dw'|}{1 - |w'|^2} = \frac{|dz|}{1 - |z'|^2}$$

$$\text{可以得到不变式: } \frac{R |dw|}{R^2 - |w - w_0|^2} = \frac{r |dz|}{r^2 - |z - z_0|^2}$$

$$\frac{|d \frac{w - w_0}{R}|}{1 - |\frac{w - w_0}{R}|^2} = \frac{|d \frac{z - z_0}{r}|}{1 - |\frac{z - z_0}{r}|^2}$$

$$\Rightarrow \frac{\frac{1}{R} |dw|}{1 - \frac{1}{R^2} |w - w_0|^2} = \frac{\frac{1}{r} |dz|}{1 - \frac{1}{r^2} |z - z_0|^2}$$

$$\Rightarrow \frac{R |dw|}{R^2 - |w - w_0|^2} = \frac{r |dz|}{r^2 - |z - z_0|^2}$$

例5: 上半平面映射到圆盘, 其中把某一点 z_1 映到圆心, 这里是单位圆.

$$w = \frac{a(z - z_1)}{c(z - z_2)} = \alpha \frac{z - z_1}{z - \bar{z}_1} \quad \text{这里 } \alpha = \frac{a}{c}$$

由最大模原理, 实轴上的点在圆周上. $z = x \in \mathbb{R}$ 时, 应有 $|w| = 1$.

$$1 = |w| = \left| \alpha \frac{x - z_1}{x - \bar{z}_1} \right|$$

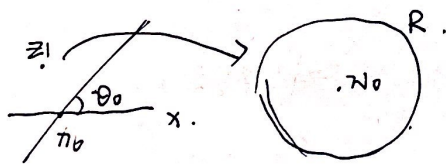
$$\text{由于 } x \in \mathbb{R}, x = \bar{x}, \text{ 故 } 1 = \left| \alpha \frac{x - z_1}{x - \bar{z}_1} \right|, \text{ 推出 } |\alpha| = 1, \text{ 故 } \alpha = e^{i\theta}, \theta \in [0, 2\pi)$$

$$\Rightarrow w = e^{i\theta} \frac{z - z_1}{z - \bar{z}_1}, \text{ 其中 } \operatorname{Im} z_1 > 0, \theta \in [0, 2\pi) \text{ (一般形式)}$$

$$\text{若令 } \theta = 0, z_1 = i, \text{ 那么 } w = \frac{z - i}{z + i} \text{ (最简单形式)}$$

$|w| = 1$ 当且仅当 $z \in \mathbb{R}$

例6. 上半平面映射到圆盘.



$$z' = (z - x_0) e^{-i\theta_0}, \quad w' = \frac{w - w_0}{R}$$

$$z_1' = (z_1 - x_0) e^{-i\theta_0}$$

$$\text{这时, } w' = \frac{z' - z_1'}{z' - \bar{z}_1'} e^{i\theta}, \theta \in [0, 2\pi)$$

$$w = w_0 + R e^{i\theta} \frac{(z - x_0) e^{-i\theta_0} - (z_1 - x_0) e^{-i\theta_0}}{(z - x_0) e^{-i\theta_0} - (\bar{z}_1 - x_0) e^{-i\theta_0}}$$

$$= w_0 + R e^{i\theta} \frac{z - x_0 - z_1 + x_0}{(z - x_0) - (\bar{z}_1 - x_0) e^{2i\theta_0}}$$

$$= w_0 + R e^{i\theta} \frac{z - z_1}{z - x_0 - (\bar{z}_1 - x_0) e^{2i\theta_0}}$$

$$= w_0 + R e^{i\theta} \frac{z - z_1}{z - x_0 - (\bar{z}_1 - x_0) e^{2i\theta_0}}, \theta \in [0, 2\pi)$$

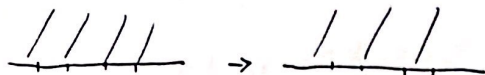
而 $\operatorname{Re} z_1 > x_0$. (?)

反解上式, 可以得到圆盘到半平面的映射.

例17. 上半平面映射到上半平面.

把边界映成边界.

$$x_k = u_k \quad (k=1, 2, 3, 4).$$



$$w = \frac{az+b}{cz+d} \Rightarrow u_k = \frac{ax_k+b}{cx_k+d} \Leftrightarrow ax_k+b = cu_k x_k + du_k, \quad k=1, 2, 3, 4.$$

形成四元齐次方程 (有无零解) 如果有解.

可以选 $a, b, c, d \in \mathbb{R}$.

由分式线性映射特征, $ad-bc \neq 0$.

结论 = 若 $a, b, c, d \in \mathbb{R}$, 则 $\text{Im} z > 0 \xrightarrow{w} \text{Im} w > 0$ 的充要条件是 $ad-bc > 0$. ($w = \frac{az+b}{cz+d}$)

证明 = 令 $z = x + iy$ 代入 $w = \frac{az+b}{cz+d}$ 中.

$$w = u + iv = \frac{acx + iay + b}{cx + d + i0} \quad a, b, c, d \in \mathbb{R}.$$

$$w = \frac{(ax+b) + iay}{(cx+d) + i0} = \frac{[(ax+b) + iay][cx+d - i0]}{(cx+d)^2 + c^2 y^2}$$

$$v = \text{Im} w = \frac{(ad-bc)y}{(cx+d)^2 + c^2 y^2} \quad \text{即}.$$

要求 $\forall y > 0, v > 0$.

$$\text{那么} \quad \frac{(ad-bc)y}{(cx+d)^2 + c^2 y^2} > 0. \quad \text{由于} ad-bc \neq 0. \quad \frac{ad-bc}{(cx+d)^2 + c^2 y^2} > 0.$$

故 $ad-bc > 0$.

• $v=0$, 当且仅当 $ad-bc y = 0$.

推论 = 如果没有 $a, b, c, d \in \mathbb{R}$, 那么结论不成立.

$$\text{e.g.} = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$$

证法2 = 令 $z = x$, $u = \frac{ax+b}{cx+d} \in \mathbb{R} \therefore a, b, c, d \in \mathbb{R}$.

这时把实轴映成实轴.

(即先证 $v=0 \Leftrightarrow w=0$).

$$\text{再对两边求导} = \frac{du}{dx} = \frac{ad-bc}{(cx+d)^2}.$$

若 $ad-bc > 0$, $\frac{du}{dx} > 0$. 单调增方向为 x 正向. (左手为上半平面).

"保向" = 内部仍为内部

考点 = 证充要条件.

□ 非分式线性映射.

$$w = e^z = e^{x+iy} = e^x (\cos y + i \sin y).$$

其中, $|w| = r = e^x > 0$.

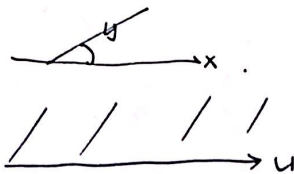
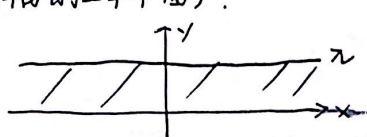
令 $y \in (0, \pi)$ 固定, e^z 对应着一条直线

$$r = e^x > 0. \quad x \rightarrow -\infty, r \rightarrow 0^+.$$

$$x \rightarrow +\infty, r \rightarrow +\infty.$$

(扫遍了没有实轴的上半平面).

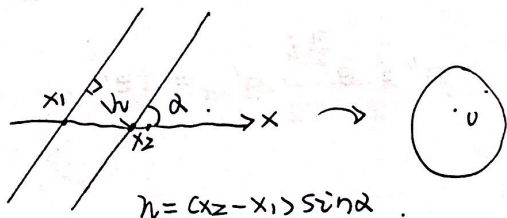
原象对应着长条:



映射 $w = e^z$.

把平行直线之间的区域 $\rightarrow$ 上半平面 $\rightarrow$ 单位圆.

例 8. 把平行长条映为单位圆盘: (给出任何一个 ρ).



$$z' = (z - x_2) e^{-i\alpha}$$

$$z'' = \frac{\pi}{h} z', \quad h = (x_2 - x_1) \sin \alpha.$$

$$= \frac{\pi}{(x_2 - x_1) \sin \alpha} (z - x_2) e^{-i\alpha}.$$

把它映成上半平面:

$$z''' = e^{z''}$$

$$\text{取 } w = \frac{z''' - i}{z''' + i}$$

$$\frac{\frac{\pi(z-x_2)e^{-i\alpha}}{(x_2-x_1)\sin\alpha} - i}{\frac{\pi(z-x_2)e^{-i\alpha}}{(x_2-x_1)\sin\alpha} + i}$$

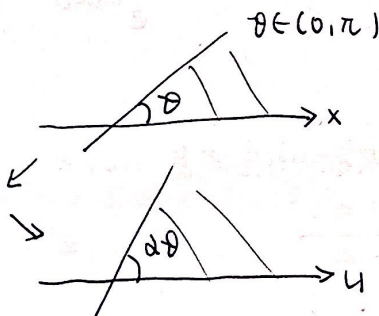
$$= \frac{\frac{\pi(z-x_2)e^{-i\alpha}}{(x_2-x_1)\sin\alpha} - i}{\frac{\pi(z-x_2)e^{-i\alpha}}{(x_2-x_1)\sin\alpha} + i} \quad (\text{一定写成这样详细形式})$$

不保角的映射:

$$w = z^\alpha, \quad \alpha > 1.$$

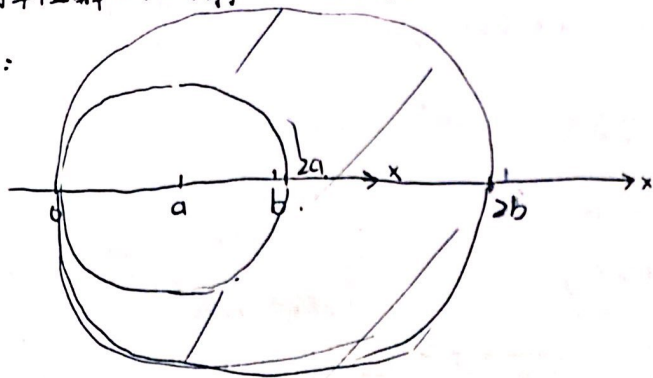
$$z = re^{i\theta}, \quad \theta \text{ 固定},$$

$$w = r^\alpha e^{i\alpha\theta}$$

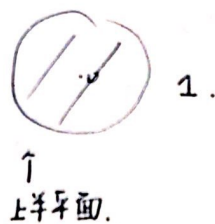


例9. 求将区域 $D = \{z \mid |z-a| > a, |z-b| < b, 0 < a < b\}$ 的单值解析映射将 D 映成 $D' = \{z \mid |z| < 1\}$

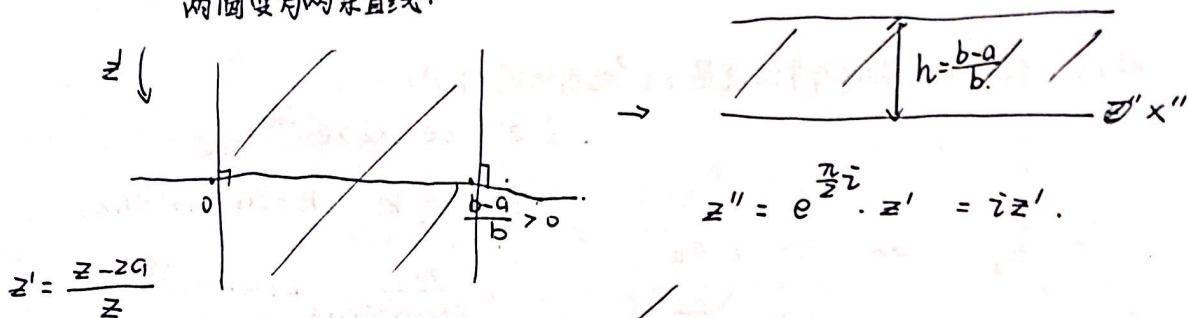
D :



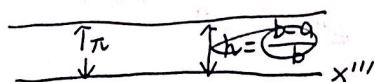
D'



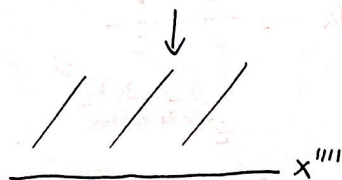
令 $z' = \frac{z-2a}{z}$, 把 $2b$ 映为正数, 把 $2a$ 映为原点.
两圆变为两条直线. $\Rightarrow ad-bc > 0$. 实轴是正向的



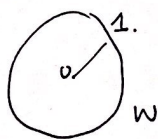
$$z'' = e^{\frac{\pi}{2}i} \cdot z' = iz'$$



$$z''' = z'' \cdot \frac{\pi}{h} = z'' \cdot \frac{\pi}{b-a} b$$



$$z'''' = e^{z'''}$$



$$w = \frac{z'''' - i}{z'''' + i} = \frac{e^{\frac{\pi b}{b-a} \cdot \frac{z-2a}{z}} - i}{e^{\frac{\pi b}{b-a} \cdot \frac{z-2a}{z}} + i} \quad (\text{保角})$$

例10.

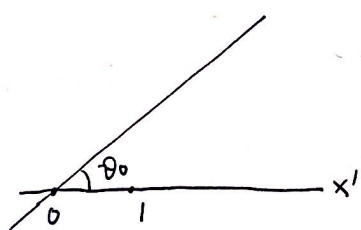
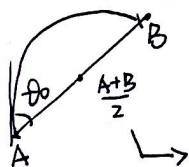
作单值解析映射为单位圆盘 $|w| < 1$.

$$0 < \theta_0 < \pi$$

$$z' = \frac{z-A}{z-B}$$

$$z'' = (z')^{\frac{\pi}{\theta_0}}$$

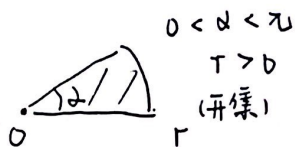
$$w = \frac{z'' - i}{z'' + i} = \frac{\left(\frac{z-A}{z-B}\right)^{\frac{\pi}{\theta_0}} - i}{\left(\frac{z-A}{z-B}\right)^{\frac{\pi}{\theta_0}} + i}$$



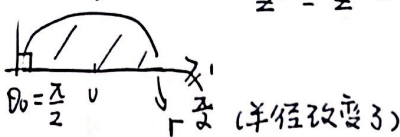
(这里不包括实轴)

例 11.

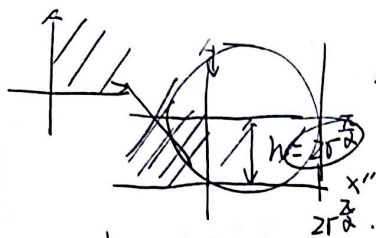
作单值解析映射, 把它映射为 $|w| < 1$. $\rightarrow$ 单位圆盘.



$$z' = z^{\frac{\pi}{\alpha}}$$



$$z'' = -\left(\frac{z' + r^{\frac{\pi}{\alpha}}}{z' - r^{\frac{\pi}{\alpha}}}\right) = \frac{z' + r^{\frac{\pi}{\alpha}}}{r^{\frac{\pi}{\alpha}} - z'}$$



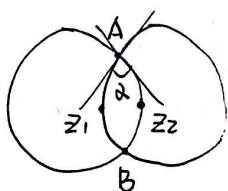
$$z''' = (z'')^2$$

$$w = \frac{z''' - i}{z''' + i} = \frac{\left(\frac{z^{\frac{\pi}{\alpha}} + r^{\frac{\pi}{\alpha}}}{r^{\frac{\pi}{\alpha}} - z^{\frac{\pi}{\alpha}}}\right)^2 - i}{\left(\frac{z^{\frac{\pi}{\alpha}} + r^{\frac{\pi}{\alpha}}}{r^{\frac{\pi}{\alpha}} - z^{\frac{\pi}{\alpha}}}\right)^2 + i}$$

考试可能指定 α .

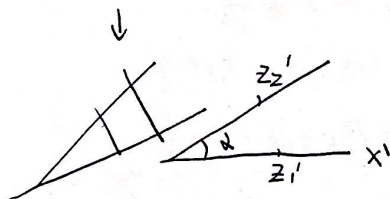
例 12.

作单值解析映射, 把它映射为 $|w| < 1$.



$$z' = \lambda \frac{z - A}{z - B}$$

$$|\lambda| = 1$$



$\Downarrow$ Riemann 定理 $\rightarrow$ 边界非 $\emptyset$.

对单连通 D 可以存在单值解析映射把它映为单位圆盘

(故可以互映)